



Zilin Pan

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SUMMARY

Master's student focused on **Embodied AI**, with experience in video and VLA/WAM modeling, physical-consistency modeling and evaluation, multi-event video generation, and large-scale robotic video data curation and preprocessing.

EDUCATION

School of Electronic and Computer Engineering, Peking University <i>Master of Computer Science and Technology</i>	2024.9 – 2027.6
School of Intelligent Systems Engineering, Sun Yat-sen University <i>Bachelor of Intelligent Science and Technology</i>	2020.9 – 2024.6

INTERNSHIP & RESEARCH EXPERIENCE

Huawei Internship — VLA/WAM Architecture and Inference Optimization	2026.6 – Present
- VLA/WAM architecture optimization using pretrained video backbones and efficient attention mechanisms. - Analysis of computational cost and inference latency across attention variants and action horizons.	
PhysisForcing: Physics Reinforced World Simulator for Robotic Manipulation	2026.3 – 2026.7
- Physics-reinforced video generation with motion-aware supervision and no inference overhead. - Pixel-level trajectory and semantic-level entity constraints with CoTracker and V-JEPA.	
Rethinking Video Foundation Model for the Embodied World (ICML 2026)	2025.9 – 2026.3
- Embodied video-model evaluation pipeline with physical-consistency and task-execution failure analysis. - Curation and depth/optical-flow preprocessing of 4M+ robotic videos for embodied model training.	
TS-Attn: Temporal-wise Separable Attention for Multi-Event Video Generation (ICLR 2026)	2025.7 – 2026.1
- A plug-and-play, training-free attention control method for multi-event video generation. - Temporal cross-attention reallocation improves event order, completeness, and video consistency.	

RESEARCH INTERESTS

Video Models	: Video generation, physical consistency modeling, and embodied video evaluation.
Embodied AI	: VLA/WAM architectures, multimodal action modeling, and inference acceleration.

TECHNICAL SKILLS

Programming & Tools:	Python, PyTorch, Hydra, Linux, Git, Codex, and Claude Code.
Modeling & Optimization:	Vision-Language-Action Models, World Action Models, and flow matching.
Robotic Data & Evaluation:	Robotic video curation, depth and optical-flow preprocessing, and evaluation.

SELECTED PUBLICATIONS

- Y. Deng*, **Z. Pan***, H. Zhang, X. Li, H. Qing, Y. Ding, Y. Zou, R. Wang, Y. Zeng, D. Zhou[†]. *Rethinking Video Foundation Model for the Embodied World*. *International Conference on Machine Learning (ICML)*, 2026. **Co-first author**.
- H. Zhang*, Y. Deng*, **Z. Pan**, P.-T. Jiang, B. Li, Q. Hou, Z. Dou, Z. Dong, D. Zhou[†]. *TS-Attn: Temporal-wise Separable Attention for Multi-Event Video Generation*. *International Conference on Learning Representations (ICLR)*, 2026.
- P. Zhang*, Y. Deng*, S. Sun*, ..., **Z. Pan**, Y. Huang, H. Liang, S. Huang, R. Zhang, E. Xie, M.-Y. Liu, D. Zhou. *PhysisForcing: Physics Reinforced World Simulator for Robotic Manipulation*. *arXiv preprint arXiv:2606.28128*, 2026.
- Z. Qiu, Z. He, Z. Zhan, **Z. Pan**, X. Xian, Z. Jin[#]. *SC-NAFSSR: Perceptual-Oriented Stereo Image Super-Resolution Using Stereo Consistency Guided NAFSSR*. *Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2023.